

Krishan Sharma

Machine Learning Engineer

krishansharma995060@gmail.com • +91 86191 64592 • LinkedIn: /in/krishan-sharma-50a89635b • GitHub: github.com/krishansharma09 • Jaipur, Rajasthan

PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with 7 months of hands-on internship experience at Dotsquares Technologies, specializing in building AI-powered applications using Python, LangChain, RAG pipelines, and Generative AI frameworks. Proven ability to design and deploy NLP systems, RESTful APIs, and document intelligence solutions. Passionate about leveraging cutting-edge LLMs and vector databases to solve real-world problems at scale.

WORK EXPERIENCE

Machine Learning Engineer Intern | [Dotsquares Technologies Pvt. Ltd.](#) | India *Aug 2024 – Feb 2025*

- Designed and deployed NLP pipelines for text classification, entity recognition, and semantic similarity, achieving over 90% model accuracy on internal benchmarks.
- Built and integrated RESTful APIs using Flask and FastAPI to serve ML model inference endpoints, reducing average response latency by 35% through async processing.
- Developed and fine-tuned machine learning models using Scikit-learn for business use cases including predictive analytics and document categorization.
- Implemented LLM-powered features using LangChain and Gemini AI, enabling intelligent document Q&A; and automated text summarization workflows.
- Collaborated with cross-functional teams in an Agile environment to deliver 3 production-ready AI features within sprint timelines.
- Optimized PostgreSQL database schemas and integrated FAISS vector stores for efficient similarity search across large-scale text corpora.

PROJECTS

AI Chatbot with Gemini + PostgreSQL + FAISS — [Python](#) · [LangChain](#) · [Gemini AI](#) · [FAISS](#) · [PostgreSQL](#) · [FastAPI](#)

- Architected a production-grade conversational AI chatbot leveraging Google Gemini LLM with LangChain orchestration for multi-turn dialogue management.
- Integrated FAISS vector store for sub-second semantic search across 10,000+ document embeddings, enabling context-aware response generation.
- Built a FastAPI backend with PostgreSQL for persistent chat history storage, session management, and user authentication — supporting 50+ concurrent users.
- Achieved 92% user satisfaction score in internal testing by combining dense retrieval with Gemini's generative reasoning capabilities.

RAG-Based Document Chatbot — [Python](#) · [LangChain](#) · [FAISS](#) · [Streamlit](#) · [OpenAI / Gemini](#)

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline allowing users to upload PDFs and query documents in natural language with high precision.
- Implemented document chunking, embedding generation, and vector indexing with FAISS, reducing irrelevant context injection by 40% vs. naive prompting.
- Deployed an interactive Streamlit UI enabling non-technical users to interact with AI-powered document search without any coding knowledge.
- Optimized prompt templates and retrieval chain logic to maintain factual accuracy and minimize hallucination in generated responses.

Binance Futures Trading Bot — [Python](#) · [Binance API](#) · [WebSocket](#) · [Pandas](#) · [NumPy](#)

- Developed an automated cryptocurrency futures trading bot using Python and Binance API, capable of executing real-time buy/sell orders based on custom algorithmic strategies.

- Implemented WebSocket-based live market data streaming for sub-second price feed ingestion, enabling low-latency trading signal generation across multiple trading pairs.
- Designed a modular strategy engine supporting technical indicators (RSI, MACD, Bollinger Bands) with configurable risk management including stop-loss and take-profit automation.
- Built a backtesting framework using Pandas and NumPy to simulate strategies on historical OHLCV data, optimizing entry/exit parameters before live deployment.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, Bash
AI / ML: Machine Learning, NLP, Deep Learning, Generative AI, Prompt Engineering
Frameworks: LangChain, Scikit-learn, TensorFlow (basics), Hugging Face Transformers
Backend: FastAPI, Flask, RESTful APIs, JWT Authentication, Pydantic
Databases: PostgreSQL, FAISS (Vector Store), SQLAlchemy ORM
Tools & Platforms: Streamlit, Gemini AI, OpenAI API, Git, GitHub, Docker (basics), Postman
Methodologies: Agile / Scrum, RAG Architecture, API Design, Model Deployment

EDUCATION

Master of Computer Applications (MCA) *Aug 2025 – Present*
JECRC University, Sitapura, Jaipur | [Currently Pursuing](#)
Bachelor of Computer Applications (BCA) *Aug 2022 – Aug 2025*
University of Rajasthan, Jaipur